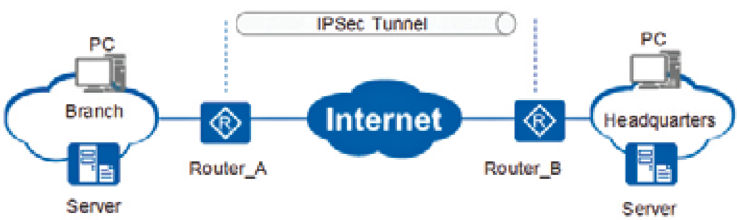
VPN Over IPsec

IPsec or Internet Protocol Security is a series of protocols that sit atop the Internet Protocol or IP layer of a system. It serves the purpose of enabling two or more hosts to exchange information securely. It does so by carrying out the encryption as well as authentication of every constituent IP packet that is part of a particular communication session.

Listed below are the sub-protocols that constitute IPsec:

- **Encapsulated Security Payload (ESP):** The ESP protocol's objective is to ensure the protection of the IP packet data from all instances of third party interference. It accomplishes this goal by encrypting the contents by making use of symmetric cryptography algorithms such as 3DES and Blowfish.
- **Authentication Header (AH):** The AH protocol shields the IP packet header from external spoofing and interference by calculating a cryptographic checksum and hashing the IP packet header fields with the aid of a secure and solid hashing function.
- **IP Payload Compression Protocol (IPComp):** The IPComp protocol is dedicated to increasing the inherent communication performance by compressing the IP payload to lessen the volume of data that is being sent.



IPSec Tunnel Diagram

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Depending on the environment, these protocols may either be invoked individually or in combination. Two modes of operations are supported by IPsec – Transport Mode and Tunnel Mode. While the former allows for secure communication between two hosts, the latter is used to establish virtual tunnels, commonly referred to as Virtual Private Networks or VPNs

OpenSSH

It is a series of network connectivity tools that are utilised to ensure secure access to remote machines. Further, SSH connections can also be used to